Guessing remainders and sign patterns in the six-point Riemann–Roch / 2-adic chain system

2026-10-15/16 (UTC) · follow-up to `REPORT.md` and `HYBRID_REPORT.md` · experiment log (observations; no hypothesis status changed). Full log: `GUESSING_REPORT.md`; table `guessing_runs.csv` (67 msolve runs, `mult` arm, $p = 268496897$, seed 1).

Strategies

- **Remainder coefficients.** Fix g coefficients of the last chain remainders $R_L = X^{2^{L+1}} \bmod q$. $g = 6$ fixes R_L; $g = 12$ fixes R_L and R_{L-1}. The guess space is p^g.
- **One-level sign split.** On each root of q, $X^{2^{k-1}} = \pm s$ with $s^2 = c$. Guess $m = \#\{+\}$ (7 options). Then $q = q_+ q_-$, with $q_\pm \mid X^{2^{k-1}} \mp s$ and each factor checked by a depth-$(k-1)$ chain.
- **Two-level sign split.** $X^{2^{k-2}}$ is one of the four fourth roots y_j of c. Guess a composition $(m_0, \ldots, m_3)$ of 6 (≤ 84 options; 68 at $k = 4$). The factors $f_j \mid X^{2^{k-2}} - y_j$ are nested inside the one-level split, so every equation stays quadratic.

Solver: msolve 0.10.1 F4, one job at a time, 1200–1800 s per system; 8.6 GB `ulimit` after the container restarts. Planted instances check a witness for the true guess.

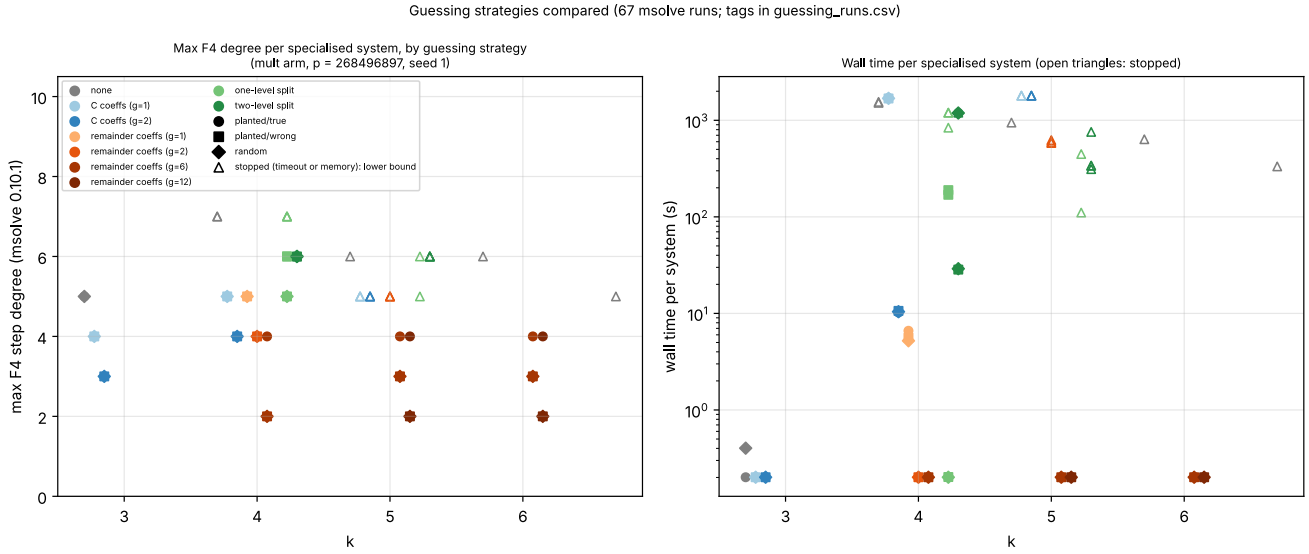


Figure 1: Max F4 degree (left) and wall time (right) per specialised system, by strategy. Filled = complete; open triangle = stopped (timeout or memory), a lower bound. Includes the earlier no-guess and C-coefficient rows. Tags in `guessing_runs.csv`.

Results (measured): max F4 degree

strategy	guess space	k = 4	k = 5	k = 6
none	—	≥ 7 (memory)	≥ 6 (memory)	≥ 6 (memory)
C coeffs $g = 1$ / 2	p / p^2	5 (≈1700 s) / 4 (≈10 s)	≥ 5 / ≥ 5 (timeout)	skipped
remainder $g = 1$	p	5 (5–7 s)	not run	not run
remainder $g = 2$	p^2	4 (0.2 s)	≥ 5 (memory)	skipped
remainder $g = 6$	p^6	true 4, wrong 2	true 4, wrong 3	true 4, wrong 3
remainder $g = 12$	p^{12}	n/a	true 4, wrong 2	true 4, wrong 2
one-level split	7	true ≥ 7; wrong 5–≥ 7	≥ 5 / ≥ 6 (memory)	skipped
two-level split	≤ 84	true **6** (1190 s); wrong 6–≥ 6	all ≥ 6 (memory)	skipped

Controls: every complete correct-guess system returned 1 solution; every complete wrong-guess or random-T system returned none. Planted and random instances cost the same for a given guess shape (e.g. two-level $(1, 2, 1, 2)$ at $k = 4$: 1189.975 s vs 1189.979 s). At $k = 4$ the systems have 25 variables with no guess, 20–32 with a one-level split, 34–37 with a two-level split and 19 with remainder $g = 6$.

Derived

- With a whole remainder guessed, the degree is flat at 4 for $k = 4, 5, 6$, because q must divide the known degree-10 polynomial $R_L^2 - c$. The guess space is $p^6 = 2^{1344}$ at P-224 size, against a per-relation budget of $\approx 2^{27.8}$. Each fixed coefficient also divides the expected relation count by p.
- Guessing 1–2 remainder coefficients behaves like guessing C coefficients (degree 5 or 4 at $k = 4$, no completion at $k = 5$), but is much faster at $k = 4$.
- Sign splits are cheap guesses but do not lower the degree. The one-level true branch stays at ≥ 7. The two-level true branch completes at 6 at $k = 4$ (unguessed: ≥ 7); at $k = 5$ every two-level system ran out of memory at degree 6. Over all 68 compositions, one $k = 4$ instance would cost an estimated 10^4–10^5 s; this is extrapolated, not measured.

Inference, limitations, next checks

Inference: on these systems, the guesses that flatten the solving degree have a guess space that is a power of p. Bounded sign-pattern guesses leave the degree at or near the unguessed values and hit the same memory wall at $k = 5$. None shows a bounded way to keep the degree near 2–3. The evidence is complete runs at $k \leq 4$ ($k \leq 6$ for whole-remainder guesses) and lower bounds elsewhere. **Limitations:** one seed and one arm; representative compositions only; time and memory caps; two container restarts interrupted two runs, which were discarded. **Next:** multi-level splits combined with remainder guesses on the small factors; a ≥ 64 GB host for $k = 5$ true branches; how the number of compositions and the per-system cost combine as the number of split levels approaches k, where it becomes exhaustive search.